

algorithms that can integrate with other TensorFlow libraries easily.

There is now a more sophisticated solution available for such high workloads using Google's marquee compute engine called Cloud Tensor Processing Unit (TPU), which is specifically designed for the TensorFlow kind of machine learning model building and services. It helps to train ML models and use them for real-time business solutions like data analytics and forecasting.

TPU based computing machines are specially designed for ML services like TensorFlow or PyTorch based solutions. They don't use CPU or RAM, and have scaler unit, vector processing unit (VPU) and matrix multiplication unit (MXU) for algorithmic processing and calculation activities. They also have high-bandwidth memory (HBM) in place of RAM for memory operations.

It can be seen in Figure 1 that TensorFlow has multiple components, including core components, interfacing components and add-on libraries. Let's understand these better.

TF servable: These are computational units used for asynchronous mode of operation, streaming results and experimental APIs to perform machine learning operations.

Servable versions: These are used to maintain one more version of servable units. A TensorFlow processing application can use one or more versions for various computational units, each serving different algorithms.

Servable streams: This is an artifactory that has sequentially ordered versions of servables. During bootstrap of TensorFlow loading, servable version is used to look up from this artifactory for experimental evaluation.

TF models: For a given machine learning task, a TensorFlow model will be created by looking up a specific servable unit with a suitable version. This is further used for evaluation and processing activities.

TF loaders: This is the bootstrapping component or API, which is mainly used to handle the life cycle of servables. Using its standardised loader API, loading and unloading (bootstrap and offload) can be done.

TF sources: This is a plugin module and used to find servable from servable stream, provide servable through TF models, and provide servable when requested from client.

TF manager: This is the controlling unit of the TensorFlow application. It is used to monitor the runtime performance and tune CPU/TPU/GPU configuration to cater to request handling, and manage life cycle of servable (startup, processing, shutdown) as well as other logging and auditing activities.

TF core: This handles servable and loader together for runtime execution, and also handles life cycle of metrics to collect the usage statistics.

Batcher: This is the processing unit of the TensorFlow application, which accelerates TPU/GPU for better performance and optimised cost. It is a procedure or master plane to handle multiple requests at a time and create machine learning tasks through TF loaders.

TF API: This provides peripheral functionalities like Estimator and Keras API library usage to support machine learning activities in OCR and image based character recognition. It is available as an out-of-the-box facility, which is easy to be hooked from other

applications using an API interface.

TF abstraction layer: These are components to build neural network models that are developed by training the machine learning model. They are used for classification through loss functions, metrics for evaluation and benchmarking.

Python Core API access: This provides language control to use core Python functions for external applications like Java, GoLang and C/C++ to invoke TensorFlow functionalities.

TensorFlow usage

TensorFlow is a great tool with countless benefits. The following are some of its use cases.

- a. **Voice/sound recognition:** Deep learning has contributed a lot to voice-cum-sound recognition. Neural networks with proper input data feed can understand audio signals. Voice recognition is mostly used in the Internet of Things, automotive, security and UX/UI (user experience/user interface). There are also voice-search and voice-activated assistants of smartphones such as Apple's Siri.
- b. **Text based applications:** Text based applications such as sentiment analysis, threat detection and fraud detection are common applications backed by TensorFlow in real time. For example, Google Translate

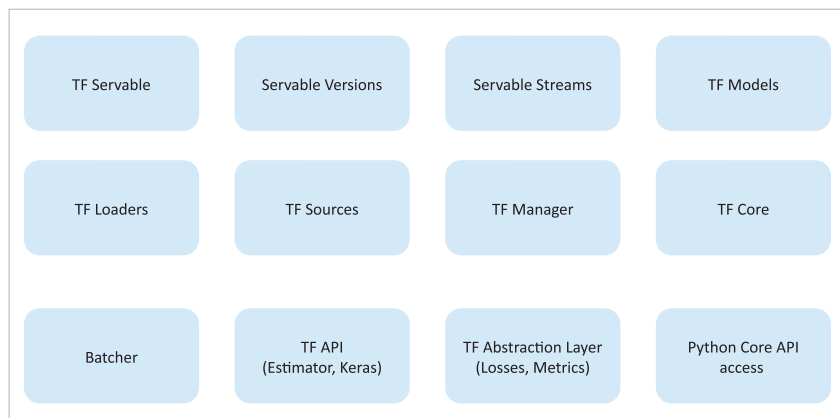


Figure 1: TensorFlow component view